Multi-Module Cognitive Swarm Architecture: Technical Briefing

Executive Summary

This document synthesizes the architectural specifications for a Distributed Cognitive Swarm Intelligence (DCSI) system, modeled as a tripartite computational unit ($N=10,000$ nodes). The architecture bridges biological grounding (Tri-Cell) with nanorobotic realism (Tri-Module), enforcing a strict **64-bit communication bottleneck** and a **64-byte memory constraint** . Central to the system is the maintenance of a **Semantic Continuity Field (SCF)** across a dynamic Riemannian manifold, allowing the swarm to achieve consensus, detect anomalies, and recover from **30% necrotic lesions** . Success is measured through a hybrid **Φ -proxy** metric, ensuring system-level coherence and functional recovery without global coordination.

1. Foundational Node Architecture: The Tripartite Model

The architecture defines each individual node as a tripartite unit, whether framed in biological (Tri-Cell) or computational (Tri-Module) terms. This separation of concerns is necessary to survive non-trivial internal dynamics and large-scale system lesions.

1.1 Core Components

Each node consists of three primary cores and two operational layers:| Core/Module | Biological Analog | Primary Function | Data/State Stored || ------ | ------ | ------ | ------ || **Core-A** | Nucleus | Identity & Regulation | Persistent hypervector identity, lineage labels, historical compression. || **Core-B** | Cytoplasm | Prediction & Metabolism | Forward model, error computation ($\hat{s} - s$), ATP/Energy management. || **Core-C** | Membrane | Interface & Gating | Receptor arrays, salience arbitration, consensus weighting, vesicle/packet I/O. |

1.2 Operational Layers

- **Processing Layer:** Executes intra-node updates, hypervector operations, and arithmetic within a 64-bit working buffer.
- **Interface Layer:** Manages packet encode/decode, neighbor caching (max 8), and the physical emission port.

1.3 State Vector Definition

A minimal defensible state-space requires 23 dimensions per node. Removing any component collapses the system's ability to maintain coherence. The vector $x_i(t)$ includes:

- **Position (p_i) and Velocity (v_i):** Continuous dynamics in a 2D/3D environment.
- **Sensory State (s_i):** Noisy local readings.
- **Hypervector (h_i):** 64-bit identity/anomaly signature.
- **Prediction Error (e_i):** Normalized mismatch between predicted and true sensory state.

- **Resource Budget (r_i):** ATP or integer energy units that deplete with computation and communication.

2. Communication Protocol: The 64-Bit Constraint

A fundamental rule of the architecture is the **extreme communication bottleneck** , mirroring biological extracellular vesicles (EVs).

2.1 Packet/Vesicle Specifications

- **Emission:** Exactly one 64-bit equivalent payload emitted per biological/clock cycle.
- **Uptake:** Each cell/node can internalize or process a maximum of **8 payloads** per cycle. Overflow is degraded or discarded.
- **Encoding:** Payloads are represented as 32-nucleotide RNA strings (2 bits/nt) or bit-packed 64-bit packets.

2.2 64-Bit Payload Schema

The 64-bit payload is designed to support consensus and lesion repair with bounded logic:| Bits | Field | Function || ------ | ------ | ------ || 0–3 | Preamble | Versioning and sync (e.g., 0b1010). || 4–11 | Identity | 8-bit lineage/identity label. || 12–15 | Message Type | 4-bit vesicle/packet class (e.g., homeostasis vs. repair). || 16–27 | Consensus Estimate | 12-bit quantized local setpoint ($\hat{s}$). || 28–39 | Stress/Error Proxy | 12-bit quantized prediction error or stress level (u). || 40–47 | Density/Location | 8-bit local density cue or spatial bucket (Morton code). || 48–51 | Flags | Bitmask for fate/repair cues (Mitosis, Apoptosis, Repair). || 52–63 | CRC/Checksum | 12-bit CRC-12 for data integrity. |

2.3 Superposition and Binding

Nodes utilize **Hyperdimensional Ternary Bipolar Sparse (HTLA)** representations.

- **Binding:** Reversible bitwise XOR or rotation-based binding (e.g., $H_{signal} = ROTL(H_{anomaly}, k) \oplus H_{location}$).
- **Superposition:** Nodes compute a **bitwise majority vote** over up to 8 received hypervectors to filter noise and reach consensus.

3. Distributed Semantic Field Dynamics

The system shifts from cryptographic hash equivalence to a continuous **Semantic Continuity Field (SCF)** model, where system health is analyzed through the dynamics of a semantic manifold.

3.1 Riemannian Manifold Interpretation

The swarm instantiates a graded Riemannian manifold (M_{HSL}). Global consensus is not a centralized artifact but the **Riemannian Center of Mass (RCM)** of all local fields.

- **Semantic Momentum (p_{sem}):** Quantifies the directional inertia of the reasoning process. High inertia represents consolidated authoritative memory; low inertia allows for contextual adaptation.

- **Coupling Model:** Modeled after high-dimensional Kuramoto oscillators. Nodes are "tethered" to the network's semantic mean via a coupling constant (K).

3.2 Semantic Drift and Divergence

Traditional hash-matching is insufficient for cognitive runtimes.

- **Pre-Hash Detection:** Systems monitor the **Jensen-Shannon (JS) Divergence** over un-sparsified probability distributions.
- **Quarantine:** Nodes exceeding a safe drift threshold ($\Delta_{sem} > \epsilon$) have their trust weight (T_i) exponentially decayed. They are forced into a "recalibration epoch" to resynchronize with the global Fréchet mean.

4. Resilience and Recovery: The Lesion Model

The architecture is explicitly designed to survive and regenerate after a **30% sudden necrotic lesion** (random removal of 30% of the population).

4.1 Lyapunov-Style Homeostatic Control

Each node implements a local Lyapunov proxy (V_i) to minimize instability:

- **Term 1:** Enforces local consensus (mismatch between local $\hat{s}$ and neighbor $\hat{s}$).
- **Term 2:** Penalizes accumulated stress (u).
- **Term 3:** Penalizes energy deficit (low ATP).
- **Term 4:** Penalizes low density (indicating a lesion edge).

4.2 Recovery Superoperators

Upon detecting a lesion (spikes in error and drops in density), the swarm triggers:

1. **Mitosis/Regrowth:** Triggered when local density is low and energy/mitosis TFs are high.
2. **Graph Rewiring:** Boundary nodes increase exploration and neighbor cache size to reconnect fragmented graph components.
3. **Redistribution:** Nodes increase the emission frequency of identity and anomaly signatures.

5. Success and Failure Criteria

The architecture is validated through quantitative, falsifiable thresholds.

5.1 Success Metrics

- **Anomaly Detection:** $\ge 85\%$ True Positive Rate; $\le 10\%$ False Positive Rate.
- **Consensus Latency:** Forms within ≤ 20 cycles (normal) or ≤ 50 cycles (post-lesion).
- **Φ -Proxy Preservation:** System-level coherence must stay ≥ 0.5 (normal) or ≥ 0.35 (immediate post-lesion), returning to ≥ 0.5 within 100 cycles.
- **Connectivity:** Giant component must include $\ge 60\%$ of nodes after a 30% lesion.

5.2 Failure Modes (Fatal Indicators)

- **Consensus Collapse:** Infinite oscillation or random noise.
- **Topology Fragmentation:** Giant component falls below 50%.
- **Hypervector Drift:** Clusters diverge rather than converging.
- **Trivial Success:** Swarm collapses to a uniform state with zero anomaly sensitivity (over-smoothing).

5.3 Essential Ablation Tests

To ensure the architecture is not "cheating" via global state, it must fail as follows when features are removed:

- **Remove HTLA Binding:** Consensus must collapse.
- **Remove Salience Scoring:** Movement must become random.
- **Reduce Swarm Size (<500):** Emergent behavior must disappear.
- **Remove Lesion Recovery Rules:** System must fail to re-cohere after node loss.

6. Operational Constraints and Energy Model

To maintain nanorobotic realism, nodes are strictly resource-constrained:

- **Memory:** Total 64 bytes per node; active working memory ≤ 32 bytes/cycle.
- **Arithmetic:** ≤ 200 integer ops, ≤ 20 floating-point ops per cycle.
- **Forbidden Operations:** Global state access, matrix inversion, multi-hop routing, and instant long-range signaling.
- **ATP/Metabolic Budget:** Mandatory costs for vesicle emission, decoding, and mitosis. If ATP reaches zero, the node becomes nonfunctional (Necrosis).